Kuncahyo Setyo Nugroho

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Summary

AI researcher and educator with 6+ years of experience in applied machine learning and AI-driven systems, including end-to-end AI pipelines from model development to deployment. Active across academic research, industry collaboration, and large-scale AI education.

Education

2023 - 2024 **Professional Engineer (Ir.) in Informatics Engineering** (3.96/4.00)
Sepuluh Nopember Institute of Technology, Indonesia
Professional Engineer Program, School of Interdisciplinary Management & Technology
Final Project: IoT-Based Livestock Management and Automated Feeding System

2020 - 2023 **Master of Computer Science (M.Kom) in Intelligent System** (3.80/4.00)
Brawijaya University, Indonesia
Department of Informatics Engineering, Faculty of Computer Science
Research Group: Intelligent System Laboratory (Interest in Affective Computing)
Thesis: Contextual Language Models for Emotion Detection in Indonesian Text

2013 - 2019 **Bachelor of Computer Science (S.Kom) in Software Engineering** (3.39/4.00)
Widya Gama University, Indonesia
Department of Informatics Engineering, Faculty of Engineering
Thesis: Complaint Text Classification System using Naive Bayes Classifier

Professional Experience

University Ambassador & Certified Instructor Jun 2024 - Present
NVIDIA Deep Learning Institute
- Delivered 15+ instructor-led workshops annually, certifying 300+ participants per year and recognized as a Platinum Tier Instructor.
- Completed 7+ certified courses on NVIDIA AI and accelerated computing technologies, with specialization recognition in AI Fundamentals and Deep Learning Applications.

AI Research Associate Nov 2023 - Present
Bioinformatics & Data Science Research Center, Bina Nusantara University, Indonesia
- Conducted applied research in AI with internal and external partners, focusing on AI applications to address real-world problems in agriculture, aquaculture, and healthcare, contributing to 5+ peer-reviewed research papers annually.
- Designed and managed AI research infrastructure at the NVIDIA-BINUS AI Research and Development Center (AIRDC), supporting large-scale model training, AI application deployment, and research data management.
- Provided mentorship in applied AI development, including supporting effective end-to-end AI implementation and hands-on proof-of-concept (PoC) solutions.
- Supervised 5+ students and junior researchers annually in AI-related projects.

Academic Coordinator of AI Mastery Program Jan 2022 - Oct 2023
PT. Orbit Ventura Indonesia (Orbit Future Academy), Indonesia
- Led the delivery of AI courses for undergraduate program for 1,250+ students across multiple cohorts.
- Designed and coordinated AI curriculum development across 10+ core topics, including learning outcomes, learning plans, instructional materials, and assessment frameworks.

– Implemented effective teaching and learning strategies in large-scale classes, while leading and coordinating 10+ mentors to improve student engagement and learning outcomes.
– Evaluated course performance and learning outcomes, driving continuous improvements in curriculum structure and instructional delivery.
– Coordinated with University partners to support and enhance the overall student learning process.

Lecture Assistant Jan 2021 - Jan 2023
Faculty of Engineering, Widya Gama University, Indonesia
– Supported the delivery of undergraduate program including Database Systems, Decision Support Systems, Data Structures and Algorithms, and Systems Analysis and Design for approximately 50+ students.
– Assisted in developing practical laboratory exercises to enhance student learning outcomes.

Laboratory Assistant of Advanced Machine Learning Feb - Jul 2021
Faculty of Computer Science, Brawijaya University, Indonesia
– Supported the delivery of an Advanced Machine Learning course for undergraduate students, involving 50+ participants, and assisted with Python programming, model implementation, and debugging during practical sessions.

Software Engineer May 2018– May 2021
ITC and Data Center, Widya Gama University, Indonesia
– Developed, tested, and maintained clean, efficient, and scalable code for internal systems.
– Participated in requirement gathering and development meetings, incorporating user feedback into software improvements.
– Assisted in debugging and resolving minor software issues, contributing to system stability and reliability.

Professional Certifications

01 – 2026 Generative AI Leader. Issued by Google Cloud
09 – 2025 Azure AI Fundamentals. Issued by Microsoft
03 – 2025 Junior Professional Engineer. Issued by The Institution of Engineers Indonesia
02 – 2025 Associate Generative AI LLMs. Issued by NVIDIA
09 – 2022 Microsoft Certified Educator. Issued by Microsoft
04 – 2018 IC3 (Internet and Computing Core Certification) Digital Literacy. Issued by Certiport

Selected Journal Papers

Hermawan, E. S., G. N. Elwirehardja, K. S. Nugroho, and B. Pardamean: *Gradient Boosted CRCNet Pipeline for Indonesian Colorectal Cancer Genotype Selection*. In: *ICIC Express Letters* 19.2 (2025), pp. 185–191. DOI: 10.24507/icicel.19.02.185.

Sulistyawan, J. S., K. S. Nugroho, and B. Pardamean: *MultiSURF-ACO: A Novel SNP Identification Approach for Colorectal Cancer Cases in Indonesia*. In: *International Journal of Fuzzy Logic and Intelligent Systems* 25.3 (2025), pp. 247–257. DOI: 10.5391/ijfis.2025.25.3.247.

Tarigan, G. A., G. N. Elwirehardja, K. S. Nugroho, and B. Pardamean: *Lighter Student Engagement Recognition in a Classroom Environment Using Skeletal Keypoints*. In: *IAENG International Journal of Computer Science* 52.6 (June 2025), pp. 1997–2014. https://www.iaeng.org/IJCS/issues_v52/issue_6/IJCS_52_6_35.pdf.

Sulistyawan, J. S., K. Julian, G. N. Elwirehardja, K. S. Nugroho, and B. Pardamean: *A Two-Step Feature Selection Approach for Identifying SNPs Associated with Colorectal Cancer*. In: *Communications in Mathematical Biology and Neuroscience* Article ID 47 (2024). DOI: 10.28919/cmbn/8468.

Nugroho, K. S., A. Y. Sukmadewa, A. Vidianto, and W. F. Mahmudy: *Effective Predictive Modelling for Coronary Artery Diseases Using Support Vector Machine*. In: *IAES International Journal of Artificial Intelligence* 11.1 (2022), pp. 345–355. DOI: 10.11591/ijai.v11.i1.pp345-355.

Nugroho, K. S., Istiadi, and F. Marisa: *Naive Bayes Classifier Optimization for Text Classification on E-government Using Particle Swarm Optimization*. In: *Jurnal Teknologi dan Sistem Komputer* 8.1 (2020), p. 21. DOI: 10.14710/jtsiskom.8.1.2020.21-26.

Selected Conference/Symposium Papers

Henry, M. M., D. C. Thio, K. S. Nugroho, and B. Pardamean: *Improving Indonesian emotion detection with openAI o4-mini text normalization*. In: *Procedia Computer Science* 269 (2025). The 10th International Conference on Computer Science and Computational Intelligence 2025, pp. 863–871. ISSN: 1877-0509. DOI: https://doi.org/10.1016/j.procs.2025.09.029.

Nugroho, K. S., S. W. Irianandra, I. Akbar, M. Isnan, and B. Pardamean: "Advancing End-To-End Indonesian Speech Emotion Recognition Using XLSR-Wav2Vec2". In: *2025 International Conference on Information Technology Research and Innovation (ICITRI)*. 2025, pp. 1–6. DOI: 10.1109/ICITRI67507.2025.11232655.

Isnan, M., G. N. Elwirehardja, K. S. Nugroho, I. L. Ritonga, R. P. Januario Gultom, E. A. Lubis, and B. Pardamean: "InDRI: Intelligent Diagnosis Radiology Images". In: *2024 6th International Conference on Cybernetics and Intelligent System (ICORIS)*. 2024, pp. 1–6. DOI: 10.1109/ICORIS63540.2024.10903684.

Sulistyawan, J. S., K. S. Nugroho, M. Isnan, and B. Pardamean: "Comparative Analysis of CNN Models for Multiclass Detection on Colon Disease". In: *2024 9th International Conference on Information Technology and Digital Applications (ICITDA)*. 2024, pp. 01–06. DOI: 10.1109/ICITDA64560.2024.10809661.

Nugroho, K. S. and F. A. Bachtiar: "Text-Based Emotion Recognition in Indonesian Tweet using BERT". In: *2021 4th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI)*. 2021, pp. 570–574. DOI: 10.1109/ISRITI54043.2021.9702838.

Nugroho, K. S., A. Y. Sukmadewa, H. Wuswilahaken DW, F. A. Bachtiar, and N. Yudistira: "BERT Fine-Tuning for Sentiment Analysis on Indonesian Mobile Apps Reviews". In: *Proceedings of the 6th International Conference on Sustainable Information Engineering and Technology*. SIET '21. Malang, Indonesia: Association for Computing Machinery, 2021, pp. 258–264. ISBN: 9781450384070. DOI: 10.1145/3479645.3479679.

Nugroho, K. S., A. Y. Sukmadewa, and N. Yudistira: "Large-Scale News Classification using BERT Language Model: Spark NLP Approach". In: *Proceedings of the 6th International Conference on Sustainable Information Engineering and Technology*. SIET '21. Malang, Indonesia: Association for Computing Machinery, 2021, pp. 240–246. ISBN: 9781450384070. DOI: 10.1145/3479645.3479658.

Books

Nugroho, K. S., M. Isnan, T. Suparyanto, and B. Pardamean: *Aplikasi Kecerdasan Buatan untuk Pertanian Modern: Klasifikasi Penyakit Daun Tanaman*. INSTIPER Press, 2025. https://ipress.instiperjogja.ac.id/book/57.

Nugroho, K. S. and F. A. Bachtiar: *Komputasi Afektif: Kemampuan Komputer untuk Mengenali Emosi Manusia*. In: *Book Chapter Seri 1 – Inovasi Teknologi dalam Digitalisasi Industri*. Vol. 1. 2021, pp. 189–203. https://s.id/pKXkF.

Nugroho, K. S. and I. Istiadi: *E-Government Cerdas Berbasis Machine Learning: Peningkatan Kualitas Pelayanan Masyarakat di Indonesia*. In: *Book Chapter Seri 1 – Inovasi Teknologi dalam Digitalisasi Industri*. Vol. 1. 2021, pp. 83–95. https://s.id/v3hld.

Intellectual Property

1. MethylBERT. Nov 27, 2023. IP No.: EC002024187405
2. InDRI: Intelligent Diagnosis of Radiology Images. Nov 23, 2023. IP No.: EC002024185948
3. Integrated Strategic System for Human Resource Management and Marketing. Sep 17, 2023. IP No.: EC002024189696
4. E-learning Integrated Expert System Using Fuzzy Inference for Disease Consultation. Nov 17, 2022. IP No.: EC00202290167
5. E-learning Expert System (ELREST). Sep 28, 2022. IP No.: EC00202269396
6. Indonesia's Leading Rabbit Farming Information System. Nov 17, 2021. IP No.: EC00202165741
7. Integration of Disease Consultation Expert System with Medical Record System. Nov 15, 2021. IP No.: EC00202164491
8. Electronic Medical Record Expert System (EMREST). Nov 15, 2021. IP No.: EC00202164510

Awards

1. Best Paper, 11th Information Technology International Seminar, 2025
2. Bast Paper, International Conference on Information Technology Research and Innovation, 2025
3. Best Paper, 6th International Conference on Sustainable Information Engineering and Technology, 2021
4. Best Presenter, 4th Conference on Innovation and Application of Science and Technology, 2021.
5. Research Intensification for Sustainable Education and Training (RISET) Scholarship, Brawijaya University, 2020

Invited Talks

12 – 2025 Why GPU Matters: Accelerate Your Data Science Research in Python. In: *Python Conference Indonesia* [Speaker Deck]

12 – 2025 Cloud-Powered Transformation: From Data to Smart Decisions with AI. In: *BNCC Workshop Tech Dive* [GitHub]

11 – 2025 AI-Powered Literature Review: Streamlining PRISMA Workflow for Researcher. In: *Faculty of Fisheries and Marine Science, Universitas Jenderal Soedirman*

07 – 2025 Navigating AI Wisely: When Technology Meets Morality. In: *Taldio AI Talk Series*

05 – 2025 Distributed Deep Learning: A Brief Introduction. In: *Laskar AI Talk Series* [Speaker Deck]

04 – 2025 AI Solutions for Transportation. In: *Public Lecture, Ministry of Transportation of Indonesia* [YouTube]

04 – 2025 AI-Powered Healthcare. In: *Microsoft Garuda*

03 – 2025 Work Smart, Not Hard #3 - Creating AI-Powered Solutions to Enhance Productivity and Public Services. In: *Microsoft Garuda* [YouTube]

11 – 2022 Maximize Society 5.0 with Artificial Intelligence. In: *Technowars 10, Universitas PGRI Kanjuruhan Malang* [Speaker Deck]

07 – 2022 Natural Language Processing Real-World Implementations. In: *Seminar Technology, Universitas PGRI Kanjuruhan Malang* [Speaker Deck]

12 – 2021 What Makes BERT Special? A Study on Emotion Detection in Text. In: *Python Conference Indonesia* [Speaker Deck]

11 – 2020 Text Preprocessing Pipeline for Bahasa using Python: Concept, Steps, Tools, and Examples. In: *Python Conference Indonesia* [Speaker Deck]

03 – 2020 Automating E-Government Services with Artificial Intelligence: Extensive use of Machine Learning. In: *Webinar of E-Government Transformation Toward Smart System* [Speaker Deck]

Invited Training & Upskilling

12 – 2025 AI for Productivity with Copilot. For: *CBM Global Indonesia*
12 – 2025 Digital Transformation: AI for Work Efficiency. For: *PT Solusi Jaya*
10 – 2025 Digital Transformation: AI for Work Efficiency. For: *PT Pegadaian Kanwil XII Surabaya*
05 – 2025 Building Ideation for AI Product & Solution. For: *PT AXA Financial Indonesia*
12 – 2023 Bringing Your Business Online: Building Websites with Google Sites. For: *Program Pengabdian Masyarakat, Universitas PGRI Adi Buana*
03 – 2021 Technical Guidance for Computer Operators. For: *Dinas Perhubungan Kabupaten Malang*

Mentoring

1. Instructor, Asah Program (Undergraduate). PT Presentologics (Dicoding), Aug – Dec 2025
2. Instructor, LaskarAI Program (Undergraduate). PT Presentologics (Dicoding), Feb – Jun 2025
3. Mentor, Software Engineering Program for Kampus Merdeka (Undergraduate). PT Revolusi Cita Edukasi (RevoU), Apr – Jun 2024
4. Mentor, Python Fundamentals & Python Pro Course (Junior High School to Senior High School). Kodland Pte.LTD (Kodland Indonesia). Jan 2024 – Jan 2025

Memberships

1. The Institution of Engineers Indonesia (PII)